Who holds government debt?

February 23, 2026

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Abstract

This paper develops a model in which the composition of public debt across domestic banks, domestic non-banks, and foreign investors is determined by endogenous borrowing limits. The government faces default constraints on non-bank creditors and convex costs of financial repression on bank-held debt. These constraints generate threshold effects: when total debt exceeds a critical level, the government exhausts non-bank borrowing capacity and allocates marginal debt to banks.

We structurally estimate the model using maximum likelihood on panel data of debt composition. The results identify distinct borrowing limits across creditor types and provide a tractable framework to interpret cross-country and time variation in who holds government debt.

Keywords Sovereign Debt; Debt Composition; Financial Repression

1 Introduction

Who holds government debt is central to macroeconomic stability. The distribution of debt across domestic banks, domestic non-banks, and foreign investors shapes default incentives, financial repression, and the transmission of fiscal shocks. Yet most models of sovereign debt focus on the total level of debt and treat its composition as exogenous.

This paper develops a simple framework in which the composition of government debt is endogenously determined by borrowing constraints and repression costs. The government faces default limits on domestic and foreign non-bank creditors, while accumulating debt in the banking sector generates convex financial repression costs. These constraints define a no-default region in debt space and imply threshold effects: beyond a critical level of total debt, the government exhausts non-bank borrowing capacity and reallocates marginal debt to banks.

We structurally estimate the model using panel data on debt composition across countries and over time. The estimation identifies distinct borrowing limits across creditor types and provides a tractable framework to interpret variation in who holds government debt.

1.1 Related literature

The literature on sovereign debt composition and financial repression is extensive. Broner, Martin, and Ventura (2010) study sovereign risk and secondary markets. This section is a placeholder; the full literature review will discuss the determinants of who holds government debt across domestic banks, non-banks, and foreign investors.

2 Model

Decompose total government debt d between components owned by banks (b) and non-banks (n), at home (h) and abroad (f)

$$d = d_n + d_b, \tag{1}$$

$$d_n = d_h + d_f. \tag{2}$$

We assume that the government does not default on home banks, because it is very costly to default on home commercial banks and it does not bring any benefit to default on the home central bank.

The government can default on home non-bank and foreign creditors, d_h and d_f. There are different assumptions one can make regarding default.

Selective default. First, one can assume that the government can default separately on home and foreign creditors.[1] Let us assume that the cost of defaulting on debt of type $\theta = h, f$ is the sum of a fixed cost, γ_θ, plus a variable component proportional to the

[1] Selective default requires market segmentation at the time of default (see Broner, Martin, and Ventura (2010)).

size of debt that is defaulted on, $\lambda_\theta d_\theta$. The benefit of defaulting is $\beta_\theta d_\theta$. The no-default constraint on debt is

$$\beta_\theta d_\theta \leq \gamma_\theta + \lambda_\theta d_\theta, \tag{3}$$

implying an upper bound on the level of debt,

$$d_\theta \leq D_\theta \quad \text{with } D_\theta \equiv \frac{\gamma_\theta}{\beta_\theta - \lambda_\theta}, \quad \theta = h, f. \tag{4}$$

On theoretical grounds, defaulting on residents should be less beneficial than defaulting on foreigners ($\beta_h \leq \beta_f$).[2] Defaulting on residents should be more costly than defaulting on foreigners because it causes more financial disruption at home ($\gamma_h \geq \gamma_f$ and $\lambda_h \geq \lambda_f$). This implies that the government should be able to sell more debt at home, $D_h \geq D_f$.

Condition (4) defines a rectangular region in the (d_h, d_f) space (see Figure 1). For a given total non-homebank government debt d_n, the debt composition can be anywhere on the segment AB. The maximum level of d_n is reached when the $d_n = d_h + d_f$ line goes through the angle C, i.e., $d_h = D_h$ and $d_f = D_f$ and the maximum value of d_n is

$$D_n = D_h + D_f. \tag{5}$$

Comprehensive default. Selective default may be impossible because of market integration or because of *pari passu* clauses in debt contracts. We now show that under some conditions, this does not lead to a very different model of the determination of the composition of debt relative to the case with selective default.

Let us assume that the government either defaults on both d_h and d_f or does not default at all. The benefit of defaulting is $\beta_h d_h + \beta_f d_f$ and the cost of defaulting is $\gamma + \lambda_h d_h + \lambda_f d_f$. Thus, the government does not default if and only if

$$\frac{d_h}{D_h} + \frac{d_f}{D_f} \leq 1, \tag{6}$$

where D_h and D_f are defined as before, setting $\gamma_h = \gamma_f = \gamma$.

Observe that the default region is no longer a rectangle but a triangle (see Figure 2). The maximum level of debt is reached when all the debt is sold at home if $D_h > D_f$. More generally, equation (5) is replaced by $D_n = \max(D_h, D_f)$.

It would make sense, however, to assume that the no-default region is *strictly* convex. For example, if there are coordination problems between home and foreign creditors, the default cost would be higher when those creditors have more equal weights in the holding of debt. The cost of defaulting could be a concave (rather than linear) function of the debt levels d_h and d_f.[3] In reduced form, convexity could be captured by assuming

[2] The sovereign debt literature typically assumes that defaulting on residents has no benefit. However, it could reduce the distortionary taxation required to repay the debt.

[3] This intuition can be formalized by allowing default costs to include an interaction term across creditor groups. Let default costs take the form $C(d_h, d_f) = \gamma + \lambda_h d_h + \lambda_f d_f + f(d_h, d_f)$, with $\partial^2 f / \partial d_h \partial d_f > 0$. Such interaction costs imply higher default costs when debt is held more evenly across home and foreign creditors, generating a strictly convex no-default region as in equation (7).

that (6) is replaced by

$$\left(\frac{d_h}{D_h}\right)^{\rho} + \left(\frac{d_f}{D_f}\right)^{\rho} \leq 1, \tag{7}$$

with $\rho \geq 1$. If there is sufficient convexity, the maximum level of debt D_n is achieved for an interior solution where both residents and foreigners are lending to the government, like in the case of selective default (Figure 2). As ρ goes to infinity we converge to the case with selective default.

What matters for our results is that there is enough convexity in the no-default region to ensure an interior solution. To simplify the analysis, we will assume that this region is a rectangle, like in the case of selective default, but selective default is not a maintained assumption.

Financial repression. The government does not default on banks but accumulating government debt in the banking sector might have costs, for example by crowding out lending to the real sector or because very large bank balance sheets complicate the operation of monetary policy or the management of financial risks.

We assume that the costs of financial repression are zero if bank-held debt d_b is below a threshold D_b. Above that threshold, financial repression has a cost $C(d_b)$ where function $C(\cdot)$ is increasing and convex.

Equilibrium. Assuming no default, for a given level of total debt $d = d_h + d_f + d_b$, the optimal debt composition minimizes $C(d_b)$ conditional on $d_h \leq D_h$ and $d_f \leq D_f$. If the cost of financial repression is very high, the government might prefer to default but we assume this possibility away as our empirical sample does not include any case of default.

The equilibrium is then easy to determine. Let us define the threshold $D \equiv D_n + D_b$. If $d \leq D$, the government can choose any debt composition $(d_\theta)_{\theta=h,f,b}$ such that $d_\theta \leq D_\theta$. If $d > D$, the debt composition is uniquely determined and given by $d_h = D_h$, $d_f = D_f$ and $d_b = d - D_n$.

3 Parameterizing the model

This section is a placeholder. The full parameterization will describe estimation of borrowing limits D_h, D_f, and D_b on panel data of debt composition. Figure 3 and Table 1 are retained from the HAFiscal template as required examples.

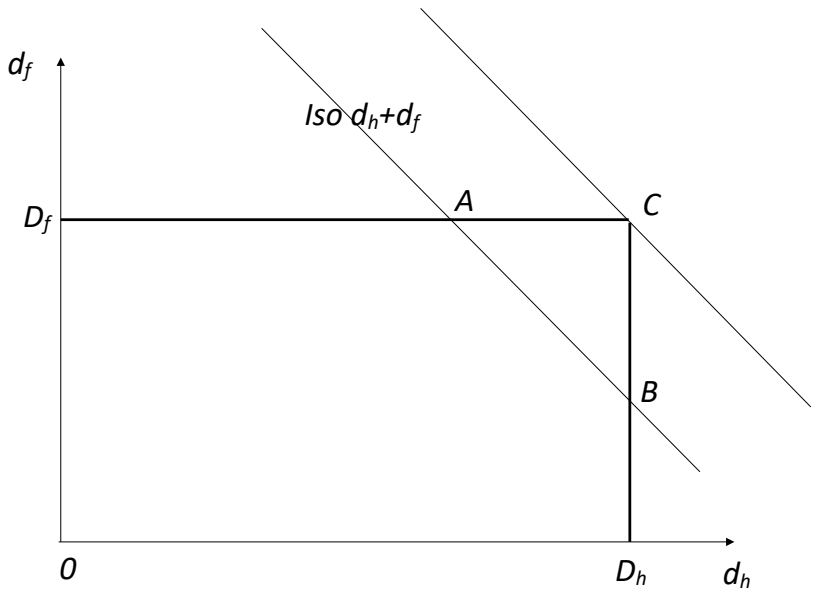


Figure 1 No default region: selective default

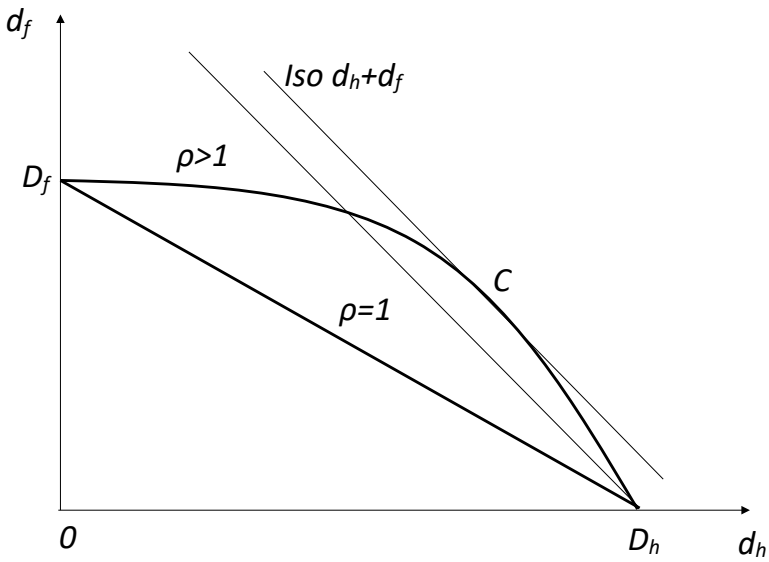
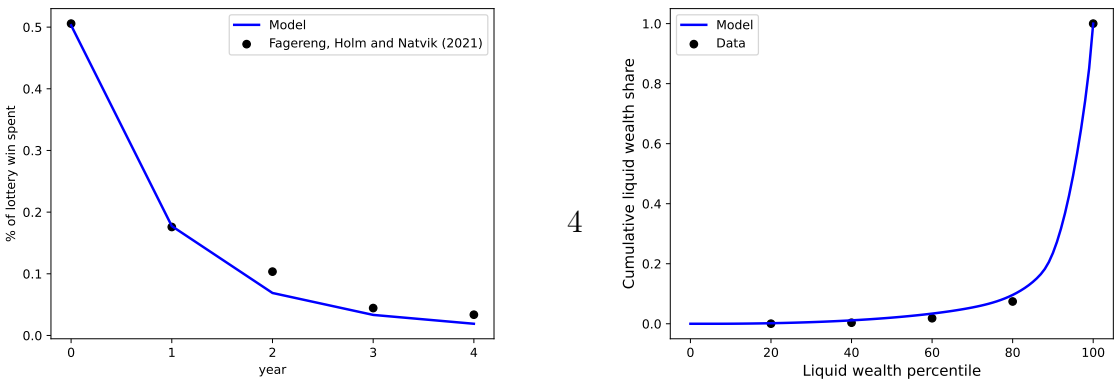


Figure 2 No default region: comprehensive default

Figure 3 Model fit to spending behavior and wealth distribution



(a) Spending dynamics after lottery win

(b) Liquid wealth distribution

Table 1 MPC's across wealth quartiles and in total

Targeting K/Y and avg MPC by quartile

	MPC					
	1st WQ	2nd WQ	3rd WQ	4th WQ	Agg	K/Y
Model	0.27	0.48	0.60	0.66	0.50	6.58
Data	0.39	0.39	0.55	0.66	0.51	6.60

Table 2 Estimated discount factor distributions and estimation targets

Panel A: Estimated discount factor distributions

	Dropout	Highschool	College
(β_e, ∇_e)	(0.719, 0.318)	(0.929, 0.072)	(0.983, 0.014)
(Min, max) in approximation	(0.447, 0.991)	(0.868, 0.991)	(0.971, 0.995)

Panel B: Estimation targets

	Dropout	Highschool	College
Median LW/ quarterly PI (data, percent)	4.64	30.2	112.8
Median LW/ quarterly PI (model, percent)	4.64	32.00	112.81

Note: Panel (A) shows the estimated parameters of the discount distributions for each education group. It also shows the minimum and maximum values we use in our discrete approximation to the uniform distribution of discount factors for each group. Panel (B) shows the weighted median ratio of liquid wealth to permanent income from the 2004 SCF and in the model. In the annual data from the SCF, the annual PI is divided by 4 to obtain a quarterly number.

Note: This figure demonstrates the calibration of the splurge factor (Section 3). Panel (a) shows the model's fit to the dynamic consumption response following lottery wins in Norway, as estimated by Fagereng, Holm, and Natvik (2021) using millions of population registry records. The splurge factor of $\varsigma = 0.249$ allows the model to match both the high initial MPC and the gradual spending over subsequent years. Panel (b) shows the model's fit to the U.S. liquid wealth distribution from the 2004 SCF, with 92.6% of liquid wealth held by the top income quintile. See Section 3 for the liquid wealth definition following Kaplan and Violante (2014).

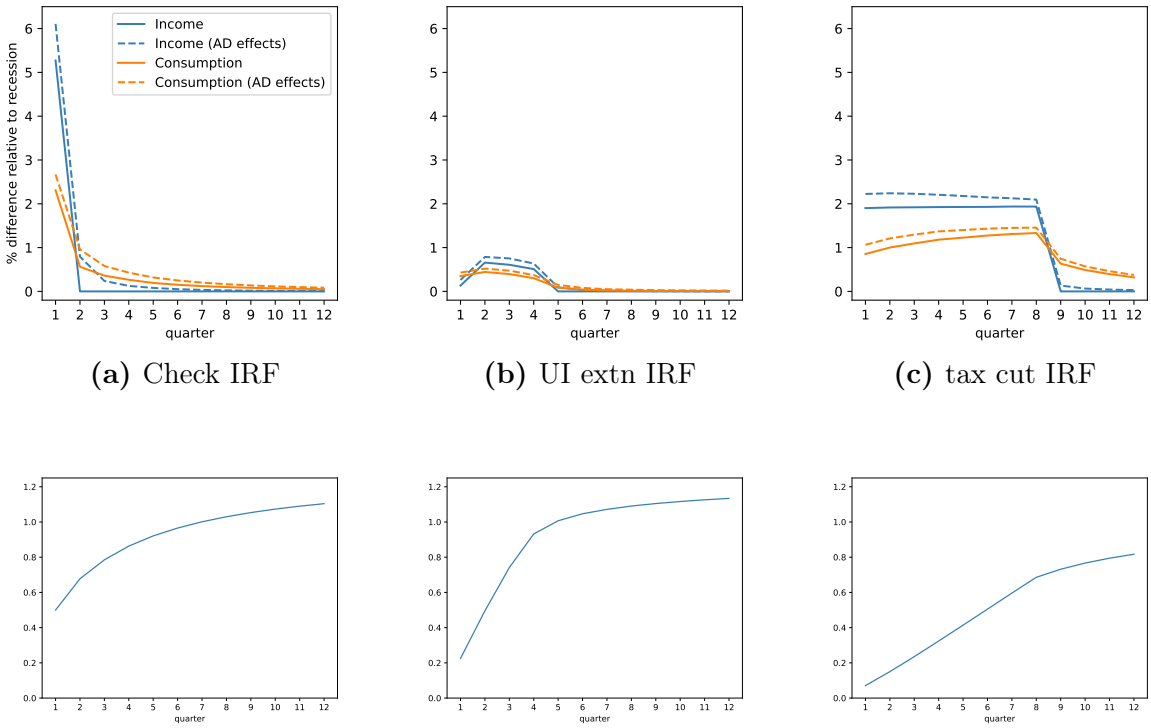
4 Comparing fiscal stimulus policies

This section is a placeholder. The cumulative multiplier is defined as

$$M(t) = \frac{NPV(t, \Delta C)}{NPV(\infty, \Delta G)}, \tag{8}$$

where ΔC is additional consumption and ΔG is government expenditure. Figure 4 and Table 3 are retained from the HAFiscal template as required examples.

Figure 4 Policy effectiveness during recessions with aggregate demand effects



(a) Check IRF **(b)** UI extn IRF **(c)** tax cut IRF

(d) Check multiplier **(e)** UI extn multiplier **(f)** tax cut multiplier

Note: This figure compares policy effectiveness during recessions (Section 2). Recession periods are characterized by higher unemployment rates and increased economic uncertainty. The model demonstrates that UI extensions become particularly effective during recessions due to better targeting of high-MPC households. Stimulus checks maintain effectiveness but with diminished relative performance compared to UI extensions. Payroll tax cuts show the least effectiveness across all economic conditions, confirming the robustness of the main policy rankings.

Table 3 Comparing fiscal stimulus policy effectiveness

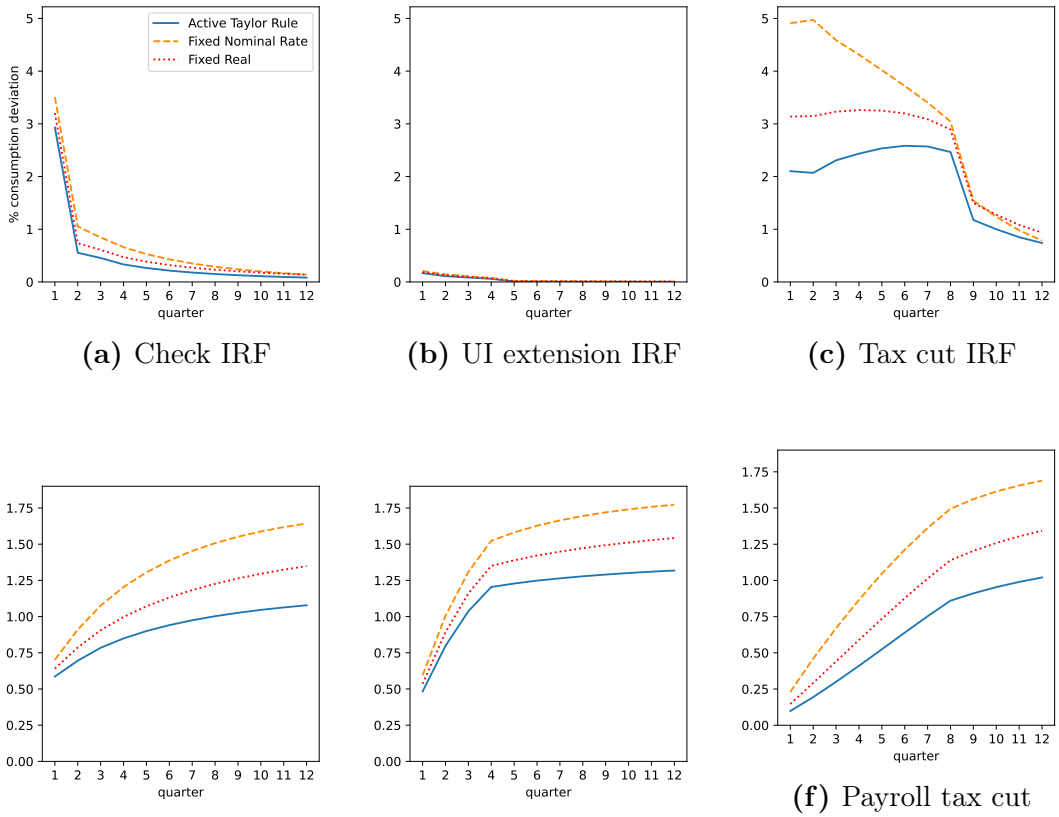
	Stimulus check	UI extension	Tax cut
10y-horizon Multiplier (no AD effect)	0.878	0.906	0.846
10y-horizon Multiplier (AD effect)	1.228	1.209	0.975
10y-horizon (1st round AD effect only)	1.153	1.147	0.949
Share of policy expenditure during recession	100.0%	79.6%	57.8 %
Share of policy cons. stimulus during recession	73.6%	81.1%	41.8 %

Note: Policies are implemented during a recession with or without the aggregate demand effect active. Multipliers show cumulative consumption response over 10 years relative to total policy cost. Share percentages indicate the proportion of policy expenditure and induced consumption stimulus occurring during the recession period. The row "1st round AD effect only" captures the direct consumption impact of the policies and the additional boost to consumption resulting from the aggregate demand effect acting on the direct consumption impact. It does not include higher-round aggregate demand effects materializing on aggregate demand effects acting on indirectly stimulated consumption.

5 Robustness in a HANK and SAM Model

This section is a placeholder. Figure 5 is retained from the HAFiscal template as a required example.

Figure 5 General equilibrium policy comparison in HANK-SAM model



(a) Check IRF **(b)** UI extension IRF **(c)** Tax cut IRF

(d) Check multiplier **(e)** UI extension multiplier **(f)** Payroll tax cut multiplier

Note: This figure presents general equilibrium results from the HANK-SAM model (Section 5). Subfigures (a)-(c) show consumption impulse responses under three monetary policy rules: active Taylor rule, fixed nominal rate (ZLB), and fixed real rate. Subfigures (d)-(f) show corresponding cumulative multipliers over time. The general equilibrium framework generates multipliers through an intertemporal Keynesian cross mechanism, particularly pronounced when monetary policy is passive. Results validate the partial equilibrium findings that UI extensions achieve the highest multipliers across all policy environments.

6 Conclusion

This paper proposes a simple model linking sovereign default risk, financial repression, and the composition of government debt. Borrowing limits on non-bank creditors and costs of bank absorption jointly determine how debt is allocated across creditor types. When total debt is sufficiently high, composition becomes constrained and shifts mechanically toward the banking sector.

Appendices

A Details of the HANK and SAM Model

This appendix is a placeholder. The final goods producer uses a CES production function:

$$Y_t = \left(\int_0^1 Y_{jt}^{\frac{\epsilon_p - 1}{\epsilon_p}} \, dj \right)^{\frac{\epsilon_p}{\epsilon_p - 1}},$$

where ϵ_p is the elasticity of substitution.

B Results in a Model Without the Splurge

Full appendix PDF available at: `GitHub: Subfiles/Appendix-NoSplurge.pdf`

Click any section title below to view it online in HTML:

B.1 Introduction

B.2 Matching the iMPCs without the splurge

B.3 Estimating discount factor distributions without the splurge

B.4 Multipliers in the absence of the splurge

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